

ISLAM EMAD AHMED ISMAIL

Mansoura, Dakahlia, Egypt • +20 155 993 3466 • islamemad811@gmail.com
[LinkedIn](#) • [GitHub](#)

Professional Summary

Analytical and detail-oriented Electronics & Communications Engineer and Project Group Leader with hands-on expertise in bare-metal embedded firmware, real-time digital signal processing (DSP), edge computing, and applied AI automation. Proven ability to architect, implement, and debug end-to-end hardware-software systems. Highly capable of resolving complex electrical and software integration anomalies, including power integrity issues, memory fragmentation, and race conditions.

Education

Horus University

Bachelor of Science in Electronics and Communications Engineering

Egypt

2021 – 2026

- Completed final academic terms and currently preparing for capstone defense.
- **Key Focus Areas:** Embedded Systems, Real-Time DSP, RF Engineering, 5G Networks, Computer Vision.

Technical Skills

- **Embedded & MCUs:** AVR-GCC, ATmega32/32A/328P, ESP32, ESP8266, STM32, Register-Level Programming, RTOS concepts.
- **Languages:** C (C99 bare-metal), C++, Python, Bash, VHDL, Verilog.
- **Protocols & I/O:** I2C, SPI, UART (up to 256000 baud), I2S DMA, PWM, ADC (10-bit sequential sampling), ISRs, H-Bridge.
- **Real-Time DSP:** Spectral Subtraction, NLMS Adaptive Filtering, FFT/PSD Estimation, Kalman Filtering, ADPCM, μ -law.
- **Edge AI & CV:** YOLO (v8/v10/v11), NCNN inference, INT8 Quantization, NEON SIMD optimization, SORT tracking.
- **Networking & RF:** UDP/Broadcast Mesh, ESP-NOW connectionless link, RF Spectral Analysis, Cisco Network Devices.
- **AI & Automation:** Gemini API (Few-Shot & Chain-of-Thought prompting), asynchronous pipelines (asyncio), Redis Streams.
- **Simulation & Tools:** Proteus VSM, Docker, Keil uVision, STM32CubeMX, Wireshark, Spectrum Analyzers, OWASP ZAP.

Featured Engineering Projects

Autonomous Drone Detection, Tracking, and Neutralization

Horus University (2025 – 2026)

Technical Lead & System Architect (Graduation Capstone Project)

Grade: A+

- Awarded **5th Place for Best Poster at the 43rd National Radio Science Conference (NRSC 2026)**.
- Designed an edge-only, hardware-in-the-loop autonomous defense countermeasure system on a Raspberry Pi 5.
- Optimized camera frame ingestion using V4L2 DMA buffers and NEON SIMD intrinsics, bypassing memory-copy overhead.
- Resolved OS latency spikes (15–20ms) by isolating CPU cores (`isolcpus` & `irqaffinity`) and assigning real-time scheduling (`SCHED_FIFO`), stabilizing timing jitter to a sub-2ms window.

ESP32 Walkie-Talkie with Adaptive Noise Cancellation

Personal Project (2025)

Digital Voice Transceiver Developer

- Developed a full-duplex, push-to-talk-free digital voice transceiver operating over connectionless ESP-NOW.
- Built an on-device DSP pipeline featuring spectral subtraction and a 32-tap NLMS adaptive filter to isolate speech.

- Eliminated audible frame-tearing glitches by designing a lock-free, atomic triple-buffer scheme in place of standard mutexes to guarantee a strict 15ms latency budget.

VoxLink — Low-Latency Mesh Voice over ESP8266

Personal Project (2025)

Mesh Network Developer

- Engineered a router-free, multi-node voice communication network using direct, loss-tolerant UDP broadcast.
- Programmed I2S DMA to pull 20ms audio frames from an INMP441 MEMS microphone into a static ring buffer.
- Resolved silent heap fragmentation freezes by completely eliminating dynamic memory allocations in the hot path, replacing them with statically sized, compile-time arrays.

Dual-Axis Solar Tracker

ATmega32A Project (2024)

Hardware & Control Developer

- Designed a bare-metal C solar tracker on ATmega32A utilizing four LDRs in a 2x2 grid behind a physical cross-divider.
- Developed a sequential 10-bit ADC sampling routine and drove dual servo motors via Timer1 Fast-PWM with error-proportional step sizes.
- Eliminated severe servo jitter by isolating the motor power rail and implementing software-based ADC averaging.

Autonomous Vulnerability Scanner (VulnScan AI)

Personal Project (2025)

SecOps Automation Developer

- Built an asynchronous vulnerability scanning pipeline using Python asyncio, Redis Streams, Nmap, and OWASP ZAP.
- Integrated Gemini API via strict, schema-enforced Chain-of-Thought JSON prompting to triage security alerts and generate PDF remediation reports.
- Resolved a 30% API initialization failure rate by implementing a robust TCP health-check backoff loop.

Publications & Research

- **Co-Author:** "Bridging the Granularity Gap: A Systematic Review of Multi-Spectral C-UAS Architectures" – Accepted, Horus University Journal of Engineering, 2026.
- **Lead Author:** Research Draft on "Real-Time AI-Based Jamming Attack Detection in 5G Networks Using Low-Cost Software Defined Radios (SDR)".

Professional Training & Experience

Information Technology Institute (ITI - Mahara-Tech)

Jul 2025 – Sep 2025

Specialized Trainee (162 Hours) – Embedded Systems Track

- Completed rigorous training in Computer Networks, Network Security, Deep Learning, and CMOS Fabrication.
- **Capstone Project:** Engineered an interrupt-driven Bluetooth Smart Home Controller on ATmega32 using USART interrupts and HC-05, validated via Proteus VSM.

Selected Certifications

- **AWS Certified Advanced Networking** – Amazon Web Services (Oct 2025)
- **Networking Devices and Initial Configuration** – Cisco Networking Academy
- **Ansible Essentials for Network Engineers & NVIDIA License System (NLS)** – NVIDIA
- **Build with AI: Masr Edition** – Google Developer Program (May 2026)
- **Backend System Design & Enterprise DevOps** – Frontend Masters
- **CMOS Fabrication Fundamentals & Deep Learning** – ITI (Mahara-Tech)